



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

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Division: A1

ASSIGNMENT No: 7

TITLE: Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable realworld examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code

reuse and reducing redundancy. Java supports single inheritance through the extends keyword.

Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Outcome:

Successfully implemented inheritance and interfaces in Java to achieve code reusability and abstraction. Applied these concepts to build real-world applications like a Shape calculator and an E-Commerce Product System.

```
interface Sound{
    void makeSound();
}

class Animal{
    String name;
    Animal (String n){
        this.name = n;
    }
    void eat(){
        System.out.println(name + " is eating.");
    }
}

class Dog extends Animal implements Sound{
    Dog(String n){
        super(n);
    }
    public void makeSound(){
        System.out.println(name + " says: Woof!");
    }
}

public class JAVA_EXP_7{

    public static void main(String[] args) {
        Dog dog = new Dog("Tommy");
        dog.eat();
        dog.makeSound();
    }
}
```

Output:

```
C:\Users\Aashray\OneDrive\Desktop\JAVA_PRACTICALS>java JAVA_EXP_7.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Tommy is eating.
Tommy says: Woof!
```

Conclusion:

Therefore, from the experiment we got to learn and demonstrate inheritance and demonstrate using suitable real world examples

Exercise

1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

Source Code:

```
1  import java.util.Scanner;
2
3  class Shape {
4      void calculateArea() {
5          System.out.println(x: "Calculating Area...");
6      }
7  }
8
9  class Circle extends Shape {
10     double radius;
11
12     Circle(double radius) {
13         this.radius = radius;
14     }
15
16     void calculateArea() {
17         double area = 3.14 * radius * radius;
18         System.out.println("Area of Circle = " + area);
19     }
20 }
21
22 class Rectangle extends Shape {
23     double length, width;
24
25     Rectangle(double length, double width) {
26         this.length = length;
27         this.width = width;
28     }
29
30     void calculateArea() {
31         double area = length * width;
32         System.out.println("Area of Rectangle = " + area);
33     }
34 }
35
36 public class assignment7_ShapeDemo {
37     public static void main(String[] args) {
38         Scanner sc = new Scanner(System.in);
```

```

39
40     System.out.print(s: "Enter radius of circle: ");
41     double r = sc.nextDouble();
42
43     System.out.print(s: "Enter length of rectangle: ");
44     double l = sc.nextDouble();
45
46     System.out.print(s: "Enter width of rectangle: ");
47     double w = sc.nextDouble();
48
49     Circle c = new Circle(r);
50     Rectangle rect = new Rectangle(l, w);
51
52     c.calculateArea();
53     rect.calculateArea();
54
55     sc.close();
56 }
57 }

```

Output:

```

C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment7_ShapeDemo.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8
Enter radius of circle: 4
Enter length of rectangle: 4
Enter width of rectangle: 2
Area of Circle = 50.24
Area of Rectangle = 8.0

```

2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

Source Code:

```
1  interface Product {
2      void displayDetails();
3  }
4
5  class ProductInfo {
6      int id;
7      String name;
8      double price;
9
10     ProductInfo(int id, String name, double price) {
11         this.id = id;
12         this.name = name;
13         this.price = price;
14     }
15 }
16
17 class Electronic extends ProductInfo implements Product {
18
19     Electronic(int id, String name, double price) {
20         super(id, name, price);
21     }
22
23     public void displayDetails() {
24         System.out.println(x: "\nElectronic Product");
25         System.out.println("ID: " + id);
26         System.out.println("Name: " + name);
27         System.out.println("Price: ₹" + price);
28     }
29 }
30
31 class Clothing extends ProductInfo implements Product {
32
33     Clothing(int id, String name, double price) {
34         super(id, name, price);
35     }
36
37     public void displayDetails() {
38         System.out.println(x: "\nClothing Product");
39         System.out.println("ID: " + id);
```

```

39         System.out.println("ID: " + id);
40         System.out.println("Name: " + name);
41         System.out.println("Price: ₹" + price);
42     }
43 }
44
45 class Grocery extends ProductInfo implements Product {
46
47     Grocery(int id, String name, double price) {
48         super(id, name, price);
49     }
50
51     public void displayDetails() {
52         System.out.println(x: "\nGrocery Product");
53         System.out.println("ID: " + id);
54         System.out.println("Name: " + name);
55         System.out.println("Price: ₹" + price);
56     }
57 }
58
59 public class assignment7_EcommerceDemo {
60     Run | Debug | Run main | Debug main
61     public static void main(String[] args) {
62
63         Electronic e = new Electronic(id: 101, name: "Laptop", price: 65000);
64         Clothing c = new Clothing(id: 102, name: "T-Shirt", price: 799);
65         Grocery g = new Grocery(id: 103, name: "Rice Bag", price: 1200);
66
67         e.displayDetails();
68         c.displayDetails();
69         g.displayDetails();
70     }
71 }

```

Output:

```

C:\Users\Aashray\Documents\University\3rd sem\JAVA>java assignment7_EcommerceDemo.java
Picked up JAVA_TOOL_OPTIONS: -Dstdout.encoding=UTF-8 -Dstderr.encoding=UTF-8

Electronic Product
ID: 101
Name: Laptop
Price: ₹65000.0

Clothing Product
ID: 102
Name: T-Shirt
Price: ₹799.0

Grocery Product
ID: 103
Name: Rice Bag
Price: ₹1200.0

```